

URL SHORTNER SYSTEM DESIGN

FUNCTIONAL AND NON FUNCTIONAL REQUIREMENTS

Functional Requirements

1. User submits a long URL
2. User gets unique short URL
3. Redirect User from short URL to original URL
4. User can get custom aliases
5. Analytics such as:
 - Number of clicks
 - Last accessed
 - User location

Non-Functional Requirements

1. High Availability(99%)
2. High Reliability
3. Maintainable
4. Low Latency (<100ms redirect)
5. High Scalability
6. Low Response time
7. Fault tolerant
8. Secure
9. Durable storage
10. Consistent

CAPACITY ESTIMATION

Assumptions

New URL per day-100 million

Read:Write=100:1 typically

Data storage time ~ 5 years

Per record size: Short URL ~7 bytes, Long URL ~400bytes, Metadata ~50byte, Total ~ 500 bytes

Traffic estimation

Write per second(WPS): $100,000,000 \text{ writes/day} \div 86,400 \text{ sec/day} \approx 1,160 \text{ writes/sec}$

Read Per second(RPS): 100:1 ratio $\rightarrow 1,160 \times 100 \approx 116,000 \text{ reads/sec}$

At Peak (2-3x of normal): $120,000 \times 3 \sim 360,000 \text{ reads/sec}$

Storage estimation

Storage over 5 years: $100\text{M/day} \times 365 \text{ days} \times 5 \text{ years} = 182.5 \text{ billion records}$
 $182.5\text{B} \times 500 \text{ bytes} \approx 91 \text{ TB (raw)}$

Cache memory estimation

Daily reads = $116,000/\text{sec} \times 86,400 \approx 10$ billion reads/day
20% of unique URLs generating 80% traffic: → cache 20% of daily new URLs' worth, or better, estimate based on unique hot URLs accessed per day

20% of 100M new URLs/day = 20M URLs
 $20\text{M} \times 500$ bytes ≈ 10 GB/day

Key-space Estimation

If using base62 (a-z, A-Z, 0-9 = 62 characters): $62^7 \approx 3.5$ trillion combinations

Since we need ~ 182.5 billion unique codes over 5 years, 7 characters is more than sufficient

Bandwidth Estimation

Write bandwidth: $1,160$ writes/sec $\times 500$ bytes ≈ 580 KB/sec

Read bandwidth: $116,000$ reads/sec $\times 500$ bytes ≈ 58 MB/sec

Server Estimation

If a single server handles $\sim 1,000$ - $2,000$ QPS comfortably

Read servers: $116,000 / 1,500 \approx \mathbf{\sim 80}$ servers

Write servers: $1,160 / 1,000 \approx \mathbf{2-3}$ servers

API CONTRACTS SPECIFICATION

Name	Method	Endpoint	Description
Shorten URL	POST	/api/v1/urls	Creates short key mapping from long URL.
Redirection	GET	/ {short_key}	Issues HTTP 302 Found header redirect.
Get Analytics	GET	/api/.../analytics	Retrieves aggregate links click stats.
Delete Link	DELETE	/api/v1/urls/...	Deactivates and deletes key from DB/cache.

DATABASE ENTITY-RELATIONSHIP (ER) DIAGRAM

users		
PK	user_id	UUID
	email	VARCHAR(255) [UQ]
	password_hash	VARCHAR(255)
	created_at	TIMESTAMP

urls		
PK	short_key	VARCHAR(7)
	original_url	TEXT
FK	user_id	UUID (Nullable)
	created_at	TIMESTAMP
	expires_at	TIMESTAMP (Nullable)

click_events		
PK	click_id	UUID
FK	short_key	VARCHAR(7)
	timestamp	TIMESTAMP
	ip_address	VARCHAR(45)
	country	VARCHAR(3)
	referrer	TEXT
	user_agent	TEXT

owns (1 to 0..N)

triggers (1 to 0..N)

DETAILED SYSTEM ARCHITECTURE



